CLASS XII – COMPUTER SCIENCE

Question Paper: File Handling & MySQL Database Connectivity using Python

Time: 3 Hours **Maximum Marks:** 100

General Instructions

1. The question paper contains 40 questions divided into two sections.

2. Section A: 25 MCQs × 1 mark = 25 marks.

3. Section B: 15 Programming Questions × 5 marks = 75 marks.

4. All questions are compulsory.

5. Write Python programs using appropriate functions, meaningful variable names and exception handling wherever required.

6. For MySQL questions, assume that the `mysql.connector` module is available.

7. Wherever necessary, make suitable assumptions.

SECTION A – MCQs

25 × 1 = 25 Marks

Q1. Which function is used to open a file in Python?
A. open() B. file() C. openfile() D. readfile()

Q2. Which file mode opens a file for writing and truncates its existing contents?
A. r B. a C. w D. r+

Q3. Which method reads the complete contents of a text file as a single string?
A. readline() B. read() C. readlines() D. get()

Q4. What does readlines() return?
A. A string B. A tuple C. A list of lines D. A dictionary

Q5. Which mode is used to append data to an existing file?
A. a B. w C. r D. x

Q6. Which statement is used to close an opened file?
A. file.close() B. file.end() C. file.exit() D. close(file)

Q7. Which function is used to change the current position of the file pointer?
A. tell() B. seek() C. position() D. move()

Q8. What does tell() return?
A. Size of file B. Current position of file pointer C. Number of lines D. File name

Q9. Which Python module is commonly used for binary file operations involving objects?
A. csv B. pickle C. mysql D. binary

Q10. Which function of the pickle module is used to store an object in a binary file?
A. load() B. dump() C. write() D. store()

Q11. Which module is used to work with CSV files in Python?
A. csv B. CSVFile C. database D. filecsv

Q12. Which command installs the MySQL Connector/Python package?
A. pip install mysql B. pip install mysql-connector-python C. pip install connector D. pip mysql install

Q13. Which Python statement is generally used to import MySQL Connector?
A. import mysql B. import mysql.connector C. include mysql.connector D. import connector.mysql

Q14. Which function establishes a connection with a MySQL database?
A. mysql.connector.connect() B. mysql.connector.open() C. mysql.connector.database() D. mysql.connectdb()

Q15. Which object is used to execute SQL queries in Python?
A. Connection object B. Cursor object C. Query object D. SQL object

Q16. Which method is used to execute an SQL query through a cursor?
A. run() B. execute() C. query() D. sql()

Q17. Which method retrieves all records returned by a SELECT query?
A. fetch() B. fetchone() C. fetchall() D. getall()

Q18. Which method retrieves only one record?
A. fetchone() B. fetchall() C. fetchrecord() D. getone()

Q19. Which SQL command is used to add a new record to a table?
A. ADD B. INSERT C. APPEND D. CREATE

Q20. Which SQL command is used to modify existing records?
A. CHANGE B. MODIFY C. UPDATE D. ALTER

Q21. Which SQL command removes records from a table?
A. REMOVE B. DELETE C. DROP RECORD D. CLEAR

Q22. After executing an INSERT, UPDATE, or DELETE query, which method is generally required to permanently save the changes?
A. save() B. commit() C. store() D. confirm()

Q23. Which SQL clause is used to select records satisfying a condition?
A. HAVING B. WHERE C. CHECK D. IF

Q24. Which is the correct way to parameterize an SQL query in Python?
A. cursor.execute(query, values) B. cursor.sql(query, values) C. cursor.run(query, values) D. cursor.query(query, values)

Q25. What is the primary advantage of parameterized SQL queries?
A. Faster Python execution B. Smaller database C. Safer value handling and reduced SQL-injection risk D. No database required

SECTION B – PROGRAMMING

15 × 5 = 75 Marks

FILE HANDLING

Q26. Write a Python program to create a text file named STORY.TXT and store multiple lines entered by the user. Then read the file and display: (1) total number of lines, (2) total number of words, and (3) total number of characters.

Q27. Write a Python program to read ARTICLE.TXT and display the number of uppercase letters, lowercase letters, digits, spaces and special characters.

Q28. Write a Python program to read DATA.TXT and create another file named VOWELS.TXT containing only those lines that contain at least one vowel.

Q29. Write a Python program to read POEM.TXT and count the number of occurrences of each word. Display the words along with their frequencies in alphabetical order.

Q30. Write a Python program to read SOURCE.TXT and create TARGET.TXT after removing all blank lines and lines containing only spaces.

Q31. Write a Python program using the pickle module to create a binary file STUDENT.DAT containing student records in the format [RollNo, Name, Class, Marks]. Then search for a student using Roll Number and display the complete record.

Q32. Write a Python program using pickle to create a binary file EMPLOYEE.DAT containing employee records [EmployeeID, Name, Department, Salary]. Read the file and display all employees whose salary is greater than ■50,000.

Q33. Write a Python program to create a CSV file named RESULT.CSV containing RollNo, Name, English, Mathematics and Computer. Then read the CSV file and display the names of students who have scored more than 80 in Computer.

PYTHON–MYSQL CONNECTIVITY

Assume the following MySQL database: Database: SCHOOL Table: STUDENT

RollNo	Name	Class	Section	Marks
101	Amit	12	A	85
102	Riya	12	B	92
103	Rahul	12	A	76
104	Sneha	12	B	88
105	Arjun	12	A	69

Q34. Write a Python program to connect to the MySQL database SCHOOL and display all records from the STUDENT table. Use mysql.connector, a connection object, cursor object, execute() and fetchall().

Q35. Write a Python program to accept Roll Number, Name, Class, Section and Marks from the user and insert the record into STUDENT using a parameterized SQL query.

Q36. Write a Python program to accept a Roll Number and display the corresponding student's complete record. Display an appropriate message if the Roll Number does not exist.

Q37. Write a Python program to accept a Roll Number and new marks and update the student's marks. Display the number of records updated.

Q38. Write a Python program to accept a Roll Number and delete the corresponding student record. Execute DELETE, use commit(), and display an appropriate message if no matching record exists.

Q39. Write a Python program to display: (1) students scoring more than 80, (2) students scoring between 70 and 90, and (3) the student having the highest marks. Use appropriate SQL queries.

Q40. Write a menu-driven Python program to perform: 1. Display all records 2. Insert a record 3. Search by Roll Number 4. Update Marks 5. Delete a record 6. Exit. Establish the database connection once and perform the selected

operations.

Marks Distribution

Section	Topic	Questions	Marks
A	MCQs	25 × 1	25
B	Text/Binary/CSV File Handling	8 × 5	40
B	Python–MySQL Connectivity	7 × 5	35
	Total	**40 Questions**	**100**

End of Question Paper